



School of Future Tech

Case Study Report

on

43. TubeGazer

by

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TubeGazer — Creator Intelligence Dashboard

Problem Statement

Content creators often manage multiple tasks while publishing videos, such as monitoring performance, tracking viewer retention, checking copyright issues, optimizing video quality, managing watch history, and handling cross-platform distribution. Most platforms provide these features separately, making content management inefficient and time-consuming.

TubeGazer solves this problem by providing a centralized dashboard where creators can analyze video performance, optimize content quality, monitor retention, manage playback queues, verify copyrights, and track cross-platform reach from a single interface.



Tech Stack

Frontend

- React JS 19
- Vite
- JavaScript (ES6+)
- React Router DOM
- Context API
- Custom Hooks

Styling

- CSS3
- Flexbox
- CSS Grid
- Responsive Design

State Management

- `useState`

- **useReducer**
- **useMemo**
- **useCallback**
- **Context API**

Data Storage

- **Local Storage**
- **Static Mock Data**

Development Tools

- **VS Code**
- **Git**
- **GitHub**
- **Vercel**

Component Architecture

1. App.jsx

- Root component of the application.
- Manages routing and overall layout.

2. Navbar.jsx

- Displays application branding.
- Provides navigation across sections.

3. Sidebar.jsx

- Displays quick navigation links.
- Improves user accessibility.

4. VideoStatsDashboard.jsx

Purpose:

- Displays video analytics.

Functions:

- Shows video views.
- Shows video status.
- Displays copyright information.
- Renders individual video cards.

5. VideoCard.jsx

Purpose:

- Reusable card component.

Functions:

- Displays video information.
- Shows performance statistics.
- Displays status badges.

6. WatchHistoryUndo.jsx

Purpose:

- Simulates watch history tracking.

Functions:

- Stores watch progress.
- Allows users to undo accidental skips.
- Restores previous watch positions.

7. NextVideoQueue.jsx

Purpose:

- Manages upcoming videos.

Functions:

- Reorder queue.
- Remove videos.
- Move videos up/down.

8. CopyrightChecker.jsx

Purpose:

- Verifies media ownership.

Functions:

- Checks video signatures.
- Displays copyright warnings.
- Shows safe content indicators.

9. RetentionSorter.jsx

Purpose:

- Analyzes viewer engagement.

Functions:

- Sort videos by:
 - Retention Rate
 - Views
 - Likes
- Uses `useMemo()` for optimization.

10. BestTimePredictor.jsx

Purpose:

- Recommends upload timings.

Functions:

- Displays engagement heatmap.
- Suggests best posting slots.
- Helps maximize audience reach.

11. CrossPlatformReach.jsx

Purpose:

- Tracks audiences across platforms.

Functions:

- Shows platform statistics.
- Displays engagement metrics.

- Provides content distribution tips.

12. ChannelLinkHub.jsx

Purpose:

- Centralized link management.

Functions:

- Stores channel links.
- Copy-to-clipboard functionality.
- Quick access to creator profiles.

13. QualityOptimizer.jsx

Purpose:

- Simulates video quality optimization.

Functions:

- Accepts video duration.
- Splits video into segments.
- Generates quality recommendations.
- Detects processing issues.

14. VideoProcessingSafety.jsx

Purpose:

- Handles processing failures safely.

Functions:

- Detects corrupted videos.
- Displays fallback messages.
- Prevents application crashes.

15. CreatorFeed.jsx

Purpose:

- Displays creator-related updates.

Functions:

- Fetches creator news.
- Shows platform updates.
- Displays useful content tips.

Context Architecture

AuthContext.jsx + AuthProvider.jsx

Purpose

Manages authentication globally.

Features

- Login
- Logout
- User Session Storage
- Protected Access

ChannelContext.jsx + ChannelProvider.jsx

Purpose

Manages channel information globally.

Features

- Store social links
- Share channel data across components
- Prevent prop drilling

Custom Hooks

useFetchFeed.js

Purpose

- Fetches creator feed data.

- Reusable API logic.

useVideoSearch.js

Purpose

- Handles video searching.
- Filters videos dynamically.

Features

Video Analytics Dashboard

- Track video performance.
- View engagement statistics.
- Monitor content health.

Watch History Undo

- Restore previous watch positions.
- Prevent accidental skipping.

Dynamic Video Queue

- Manage upcoming videos.
- Rearrange recommendation order.

Copyright Detection

- Verify media ownership.
- Identify copyrighted content.

Retention Analysis

- Compare viewer retention.
- Sort videos by performance.

Quality Optimization

- Segment videos for efficient playback.
- Generate quality suggestions.

Upload Time Prediction

- Recommend best publishing schedules.
- Improve audience reach.

Cross-Platform Reach

- Analyze platform-specific performance.
- View audience distribution.

Channel Link Management

- Store and share social media links.
- Copy links instantly.

Error Handling

- Graceful fallback UI.
- Prevent component failures.

Authentication System

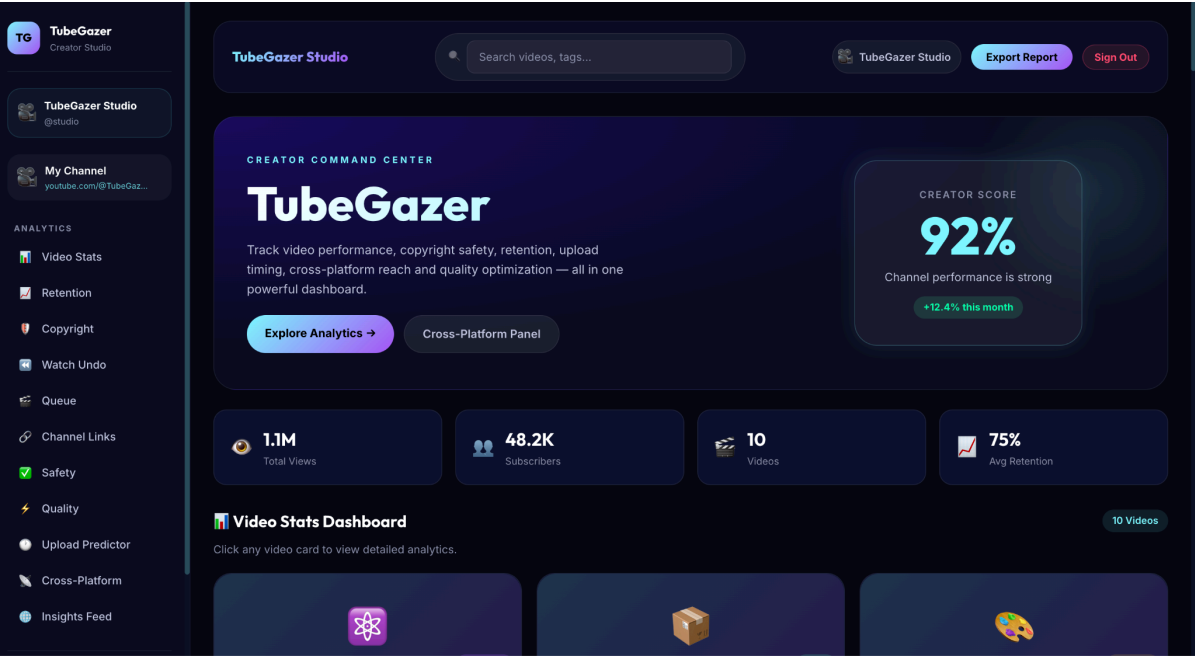
- Secure login.
- Persistent user sessions.

Responsive Design

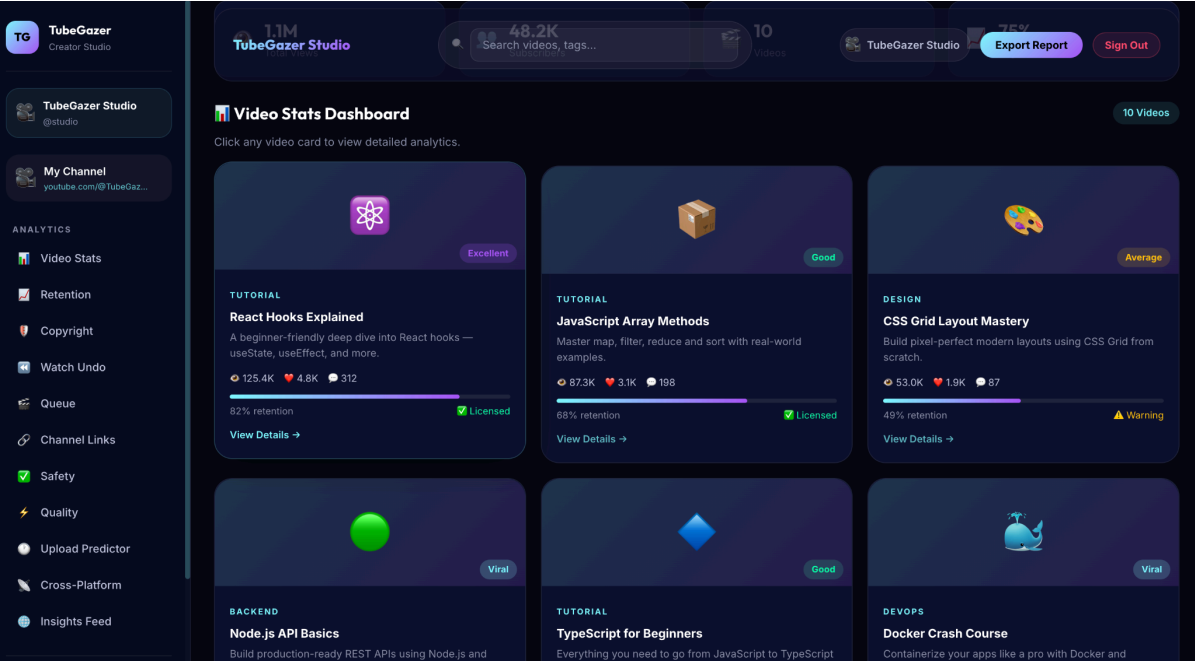
- Works across desktop and mobile devices.

Screenshots

Dashboard Overview



Video Statistics Dashboard



Copyright Checker

Copyright Checker

Verify if your video hash exists in the license database.

Enter video hash (e.g. RX101, JS202)

 **Verify Hash**

Try these hashes:

RX101

JS202

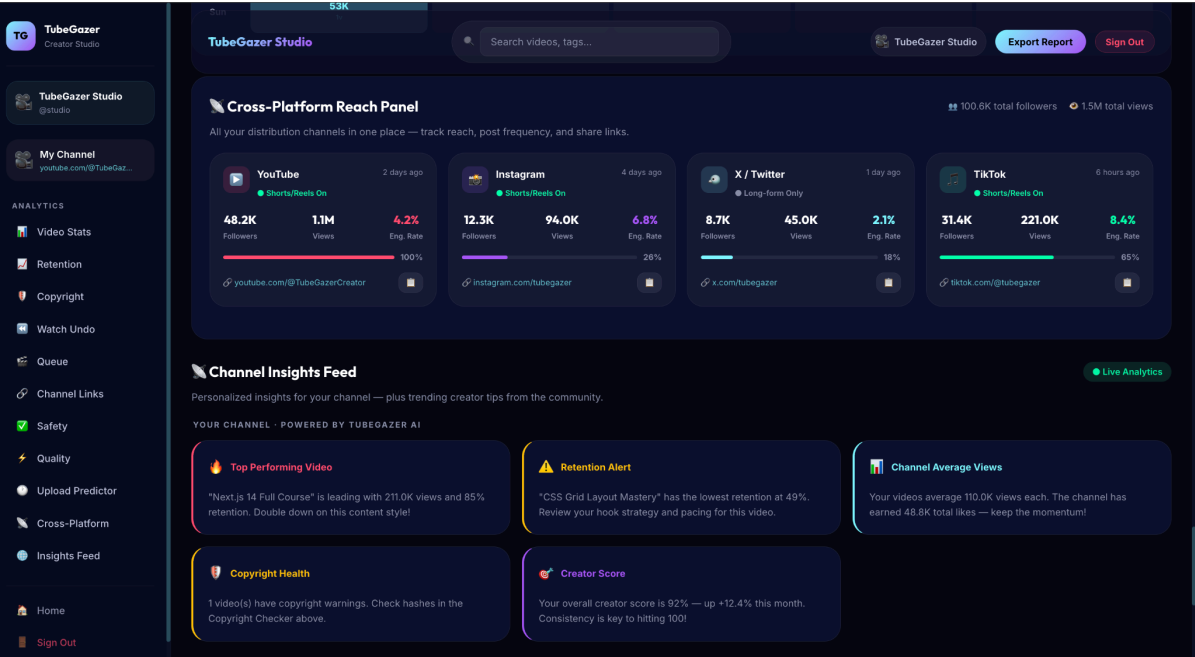
CSS303

ND404

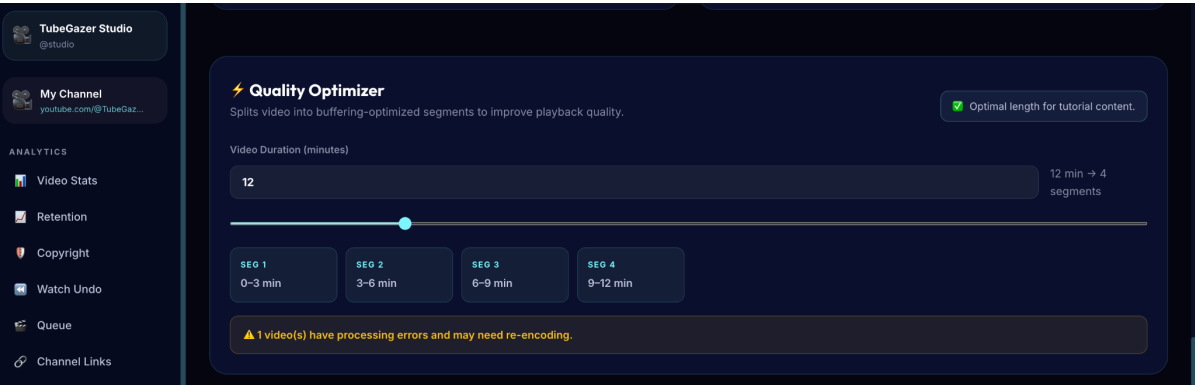
Retention Sorter



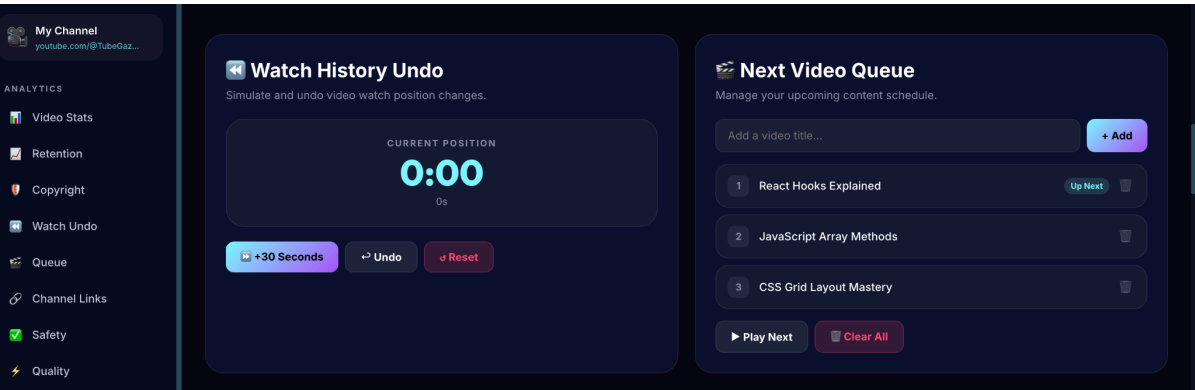
Cross Platform Reach



Quality Optimizer



Watch History Undo



Conclusion

TubeGazer is a React-based Creator Intelligence Dashboard designed to simplify content management for digital creators. The application combines analytics, retention tracking, copyright verification, video quality optimization, upload scheduling, and cross-platform reach monitoring into a single platform. By utilizing modern React concepts such as Context API, Custom Hooks, useMemo, useCallback, and React Router, the project demonstrates scalable component architecture, efficient state management, reusable code practices, and responsive UI design while providing creators with valuable insights to improve content performance.

GitHub Link: <https://github.com/irenechk/TubeGazer/tree/main>

Live Link: <https://tube-gazer.vercel.app>